ECE250: Signals and Systems

Lecture-1
August 13, 2025
GC Course Code: xwmosfls

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INFORMATION TECHNOLOGY
DELHI

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Course Outcomes

1. Students are able to classify basic signal representation; continuous time and discrete-time signals
2. Students are able to determine the response of an LTI system to continuous-time or discrete-time input signals
3. Students are able to represent both continuous-time and discrete-time periodic signals as Fourier series.
4. Students are able to analyze continuous-time signals and systems using Fourier transform and Laplace transform.
5. Students are able to analyze discrete-time signals and systems using discrete-time Fourier transform and Z-transform.

Weekly Lecture Plan

Week Number	Lecture Topic	COs Met
Week 1	Introduction to Linear Algebra; Introduction to Signals - discrete and continuous time, basic representation; Classification of signals – Continuous/ discrete; deterministic/random; even/odd, power, energy, periodic/aperiodic; singularity functions.	CO1
Week 2 and 3	System properties – Linearity (L), time-invariance (TI), causality, memory, stability; Impulse response of LTI system, Convolution (continuous and discrete time)	CO1, CO2
Week 4 and 5	Fourier series- for continuous time and discrete-time signal representation; properties of Fourier series; system transfer function and frequency domain analysis	CO1, CO2, CO3
Week 6 and 7	Continuous Time Fourier transform (FT); properties; system transfer function; analysis using FT	CO1, CO2, CO4

Week 8	Sampling and signal reconstruction	CO1, CO2, CO3
Week 9	Mid-Semester Exams	
Week 10	Discrete Time Fourier Transform (DTFT) ; properties; system transfer function; analysis using DTFT	CO1, CO2, CO4, CO5
Week 11	Laplace Transform, ROC; Properties of Laplace transform; Poles and zeros; Solution of differential equation using unilateral Laplace transform	CO1, CO2, CO4
Week 12	Mid Recess	
Week 13 and 14	Z-transform; ROC; Properties of Z-transform; Poles and zeros; System analysis	CO1, CO2, CO5
Week 15 and 16	Revision or coverage for spill over if any.	

Assessment Plan		
Type of Evaluation	**% Contribution in Grade**	
Attendance (class)	10	For Lectures
Attendance (Tutorial)	5	For Tutorials
Quiz	16	Best 8 out of 10
Assignments	10	2 assignments (5 marks each)
Mid-sem	24	
End-sem	35	

Text Book: Signals and Systems, 2nd Ed. by Alan V Oppenheim, Alan S Willsky and S. H. Nawab

- **Assessment**

Minimum overall score (absolute) required for A grade- > 80%

Minimum passing criterion:

a) At least 30% marks in the midsem component (of 24 marks)

b) At least 30% marks in the endsem component (of 35 marks)

c) At least 30% marks overall in the total quiz component (of 16 marks)

d) Overall, at least 30% marks

All the above from (a) to (d) are required to be met to pass the course.

- **Attendance:** will be taken in all of the classes- *The classroom doors will close 5 minutes after the entry of the Instructor.*
- *Please bring your Copy and pen to the classroom and do it. Students not interested in writing in the classroom will be asked to leave and will not be given any attendance.*
- Some online resources

  https://onlinecourses.nptel.ac.in/noc21_ee28/preview

  https://archive.nptel.ac.in/courses/108/104/108104100/

  https://ocw.mit.edu/courses/res-6-007-signals-and-systems-spring-2011/resources/lecture-1-introduction/
- **Re-grading of exams:** For re-grading, student is to write the question to be re-graded on the first page of the answer copy.

Plagiarism: Please check IIIT-D policy on the website.

Proxy on attendance will lead to an F grade.

Office hours: will be announced at Classroom page